Name

Mathematics Stage 5 Core

Coordinate Geometry

Book 1 Find the midpoint, gradient, and distance between two points on the Cartesian plane

Version: 261006

Feedback: https://MrDingMaths.com

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Syllabus Content

MA4-LIN-C-01 determines the midpoint, gradient and length of an interval, and graphs linear relationships, with and without digital tools

Find the midpoint and gradient of a line segment (interval) on the Cartesian plane

- Plot and join 2 points to form an interval on the Cartesian plane and use the interval as the hypotenuse of a right-angled triangle
- Apply the relationship gradient $m = \frac{rise}{run}$ to find the gradient/slope of the interval joining the 2 points
- Distinguish between intervals with positive and negative gradients from a diagram
- Explain why horizontal intervals have a gradient of 0 and vertical intervals have undefined gradients using the gradient relationship
- Determine the midpoint of horizontal and vertical intervals on the Cartesian plane
- Apply the process for calculating the mean to find the midpoint, M of the interval joining 2 points on the Cartesian plane
- Use graphing applications to find the midpoint and gradient/slope of an interval

Find the distance between 2 points located on the Cartesian plane

- Use the interval between 2 points as the hypotenuse of a right-angled triangle on the Cartesian plane and apply Pythagoras' theorem to determine the length of the interval joining the 2 points
- Use graphing applications to find the distance between 2 points on the Cartesian plane

Gradient

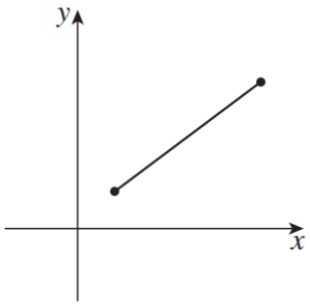
Gradient

The **gradient** measures a line's steepness.

Gradient: https://www.geogebra.org/m/jjqsdqxr

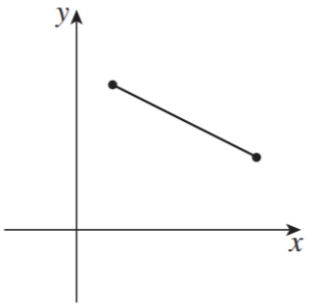
Positive Gradient

Line slopes up from left to right.

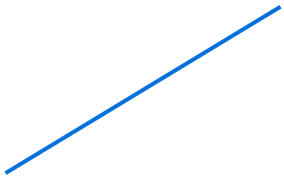


Negative Gradient

Line slopes down from left to right.



Identify positive and negative gradients

a  Positive | Negative	**b**  Positive | Negative	**c**  Positive | Negative
d Positive | Negative	**e**  Positive | Negative	**f**  Positive | Negative
g  Positive | Negative	**h** Positive | Negative	**i**  Positive | Negative

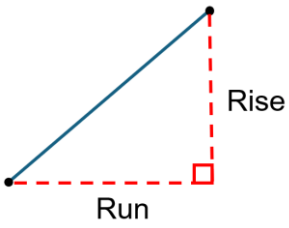
Key Ideas

1. Gradient is …… if the line moves upwards from left to right, and negative if it moves downwards from …… to ……

Gradient Formula

Gradient, m, is the ratio of **rise** to **run**.
Gradient shows **how far up or down for each unit to the right**.

$$m = \frac{\text{rise}}{\text{run}}$$



To calculate gradient:

1. Find the **rise**: vertical change between two points, from left to right.
   a. Positive if line slopes upwards from left to right.
   b. Negative if line slopes downwards from left to right.
2. Find the **run**: horizontal change between two points, moving from left to right.
3. Calculate gradient $m = \frac{\text{rise}}{\text{run}}$.
4. Simplify the fraction, if you can.

Calculate gradient of an interval

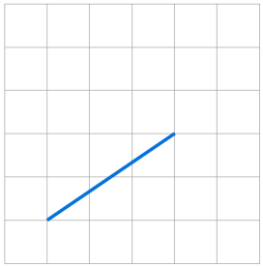
a

Rise: ……

Run: ……

$m = \frac{\text{rise}}{\text{run}}$

$=$



b

Rise: ……

Run: ……

$m = \frac{\text{rise}}{\text{run}}$

$=$



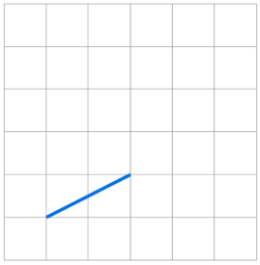
c

Rise: ……

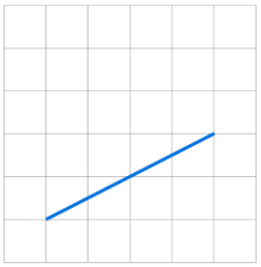
Run: ……

$m = \frac{\text{rise}}{\text{run}}$

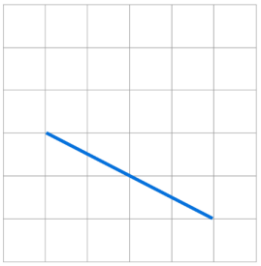
$=$



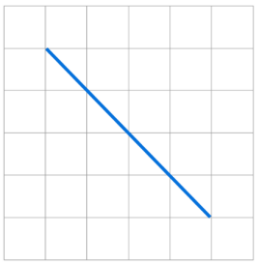
d



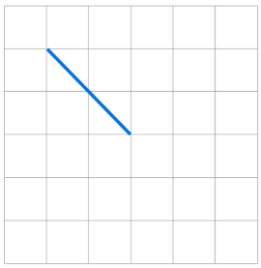
e



f



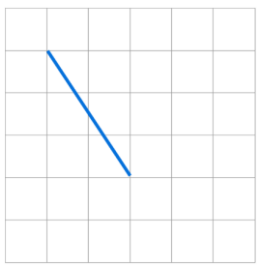
g



h



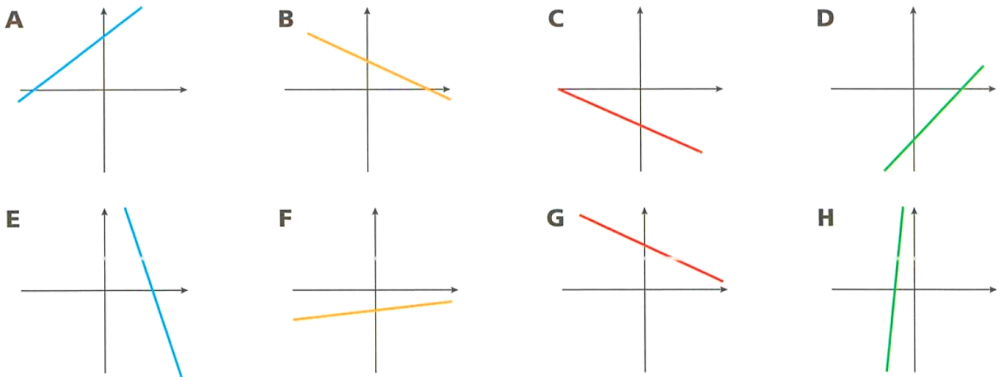
i



Key Ideas

1. Gradient is a r…… that describes the …… of a line.
2. The gradient is a number that represents how much a line moves …… for every 1 unit to the right.

1 Consider these lines. Circle the graphs with a negative gradient.



B, C, E, G

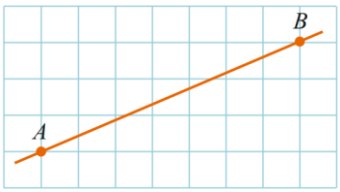
2

a Determine the gradient of the interval AB

Rise: .....

Run: .....

$$m = \frac{\text{rise}}{\text{run}} = \frac{\square}{\square}$$



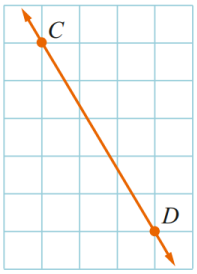
$\frac{3}{7}$

b Determine the gradient of the interval CD

Rise: .....

Run: .....

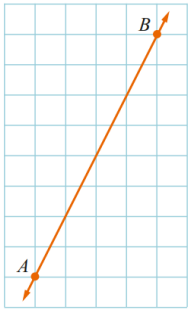
$$m = \frac{\text{rise}}{\text{run}} = \frac{\square}{\square}$$



$-\frac{5}{3}$

3 Determine whether the gradient is positive or negative and then find the gradient.

a

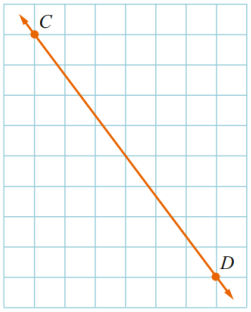


Positive | Negative

$$m = \frac{\text{rise}}{\text{run}} =$$

$\frac{8}{4} = 2$

b

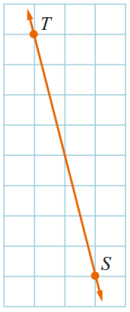


Positive | Negative

$m =$

$-\frac{8}{6} = -\frac{4}{3}$

c

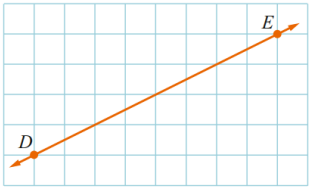


Positive | Negative

$m =$

$-\frac{8}{2} = -4$

d

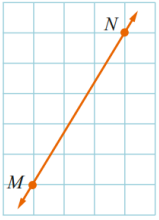


Positive | Negative

$m =$

$\frac{4}{8} = \frac{1}{2}$

e

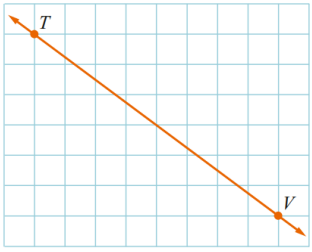


Positive | Negative

$m =$

$\frac{5}{3}$

f



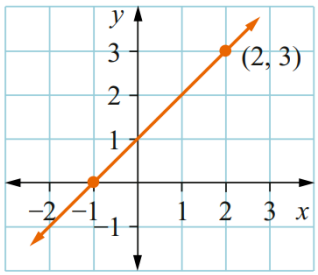
Positive | Negative

$m =$

$-\frac{6}{8} = -\frac{3}{4}$

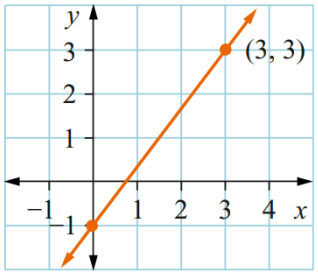
4 Find the gradient of these lines. Remember to check positive or negative.

a



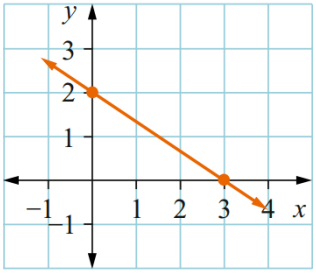
$m = 1$

b



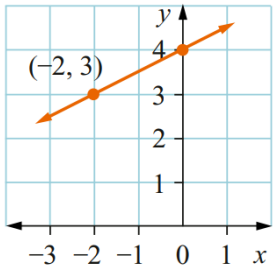
$m = \frac{4}{3}$

c



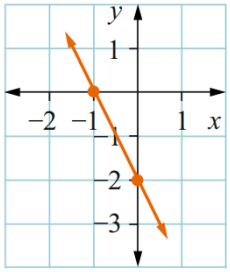
$m = -\frac{2}{3}$

d



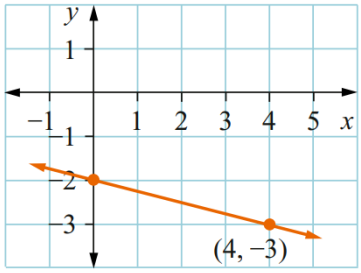
$m = \frac{1}{2}$

e



$m = -2$

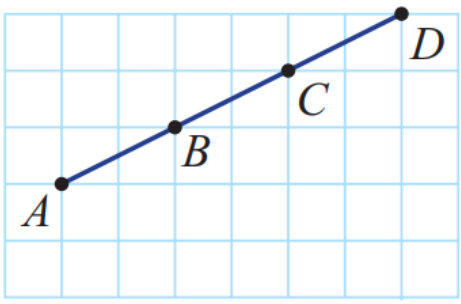
f



$m = -\frac{1}{4}$

5 Calculate the gradient of these line segments.

Interval	Rise	Run	Gradient
AB	1	2	$\frac{1}{2}$
AC	2	4	$\frac{1}{2}$
AD	3	6	$\frac{1}{2}$
BC	1	2	$\frac{1}{2}$
BD	2	4	$\frac{1}{2}$
CD	1	2	$\frac{1}{2}$



What do you notice about the gradient between points on the same line?

They are the same

Does it matter which two points you pick on a line to calculate gradient?

No

6 Tom climbs a rocky slope that rises 12 m for each 6 metres across.
His friend Jonathan climbs a nearby grassy slope that rises 23 m for each 12 m across.
Which slope is steeper?

Tom: $\frac{12}{6} = 2$, Johnathan: $\frac{23}{12} < 2$

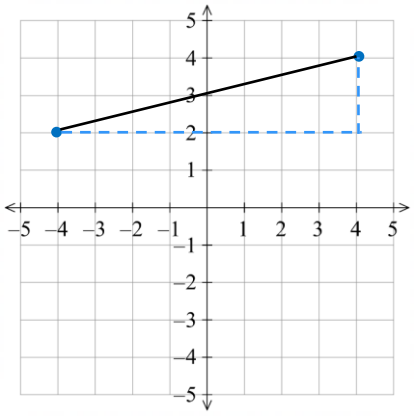
Tom's slope

Gradient from Coordinates

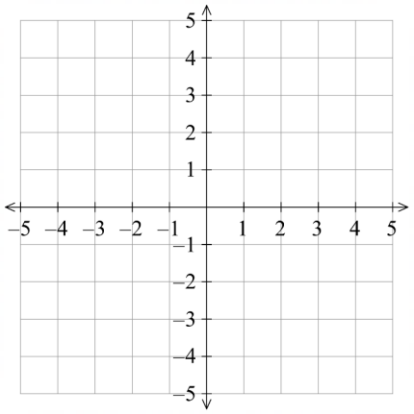
Review

☐ Plot points on the Cartesian plane and join them to form an interval

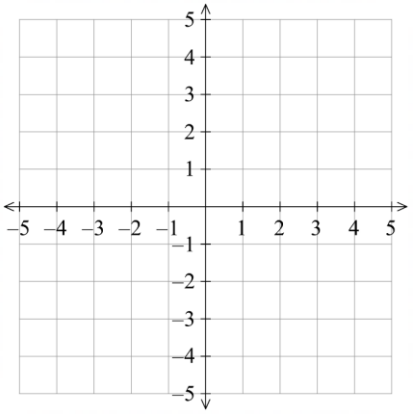
a $(-4,2)$ and $(4,4)$



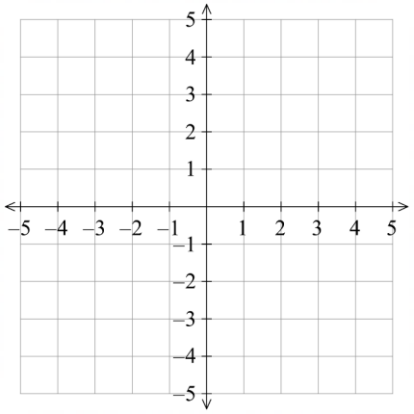
b $(0,2)$ and $(2,0)$



c $(2,3)$ and $(5,-4)$.



d $(-4,3)$ and $(1,-2)$

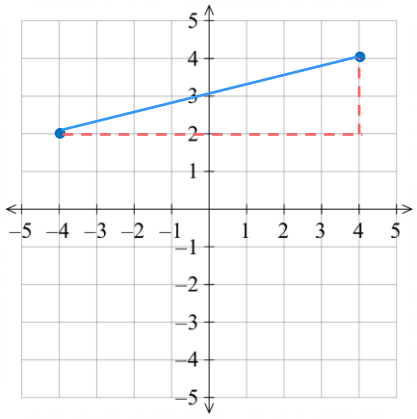


Gradient from Coordinates

1. Plot the two coordinates on the Cartesian plane.
2. Find the **rise**
3. Find the **run**
4. Calculate gradient $m = \frac{\text{rise}}{\text{run}}$.
5. Simplify the fraction, if you can.

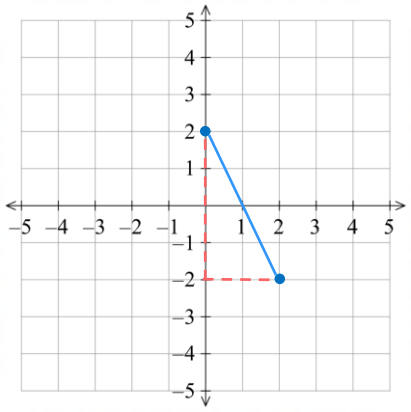
Example Calculate gradient from two coordinates by plotting the points

Find the gradient between $(-4,2)$ and $(4,4)$



1. Rise $= 2$
2. Run $= 8$
3. $m = \frac{\text{Rise}}{\text{Run}}$
   $= \frac{2}{8}$
4. $= \frac{1}{4}$

Find the gradient between $(0,2)$ and $(2,-2)$

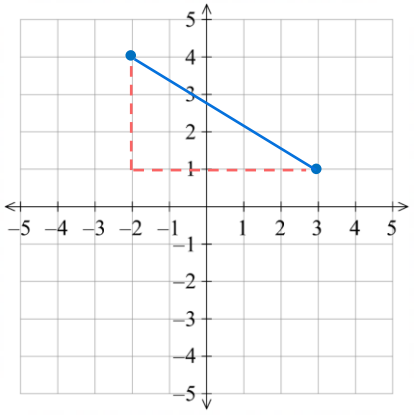


1. Rise $= -4$
2. Run $= 2$
3. $m = \frac{\text{Rise}}{\text{Run}}$
   $= \frac{-4}{2}$
4. $= -2$

Guided Practice

Find the gradient of these intervals

a $(-2,4)$ and $(3,1)$.



Rise $=$

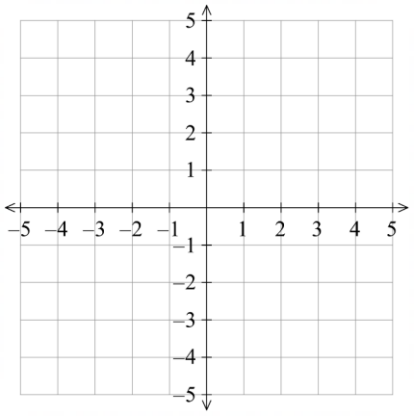
Run $=$

$m = \frac{\text{Rise}}{\text{Run}}$

$=$

$-\frac{3}{5}$

b $(0,5)$ and $(5,0)$



Rise $=$

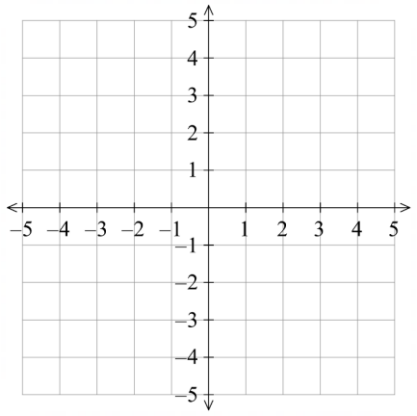
Run $=$

$m =$

$=$

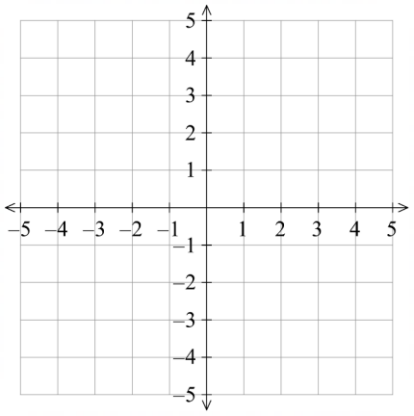
-1

c $(-3,2)$ and $(4,2)$



0

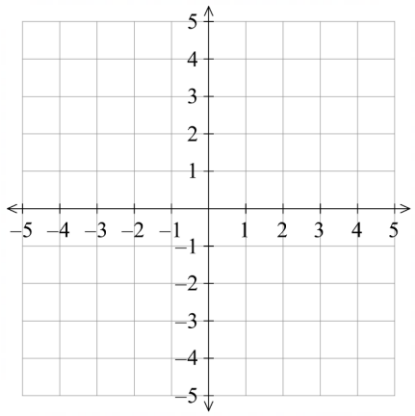
d $(-3,-1)$ and $(1,-3)$



$-\frac{1}{2}$

1 Plot these points on the Cartesian plane and calculate the gradient of the interval.

a $A(2,3)$ and $B(3,5)$



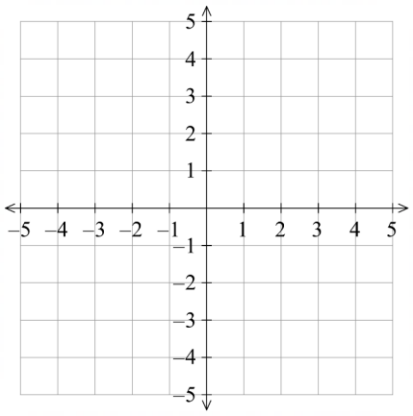
Rise:.....

Run: .....

$m =$ .....

2

b $A(0,3)$ and $B(3,0)$



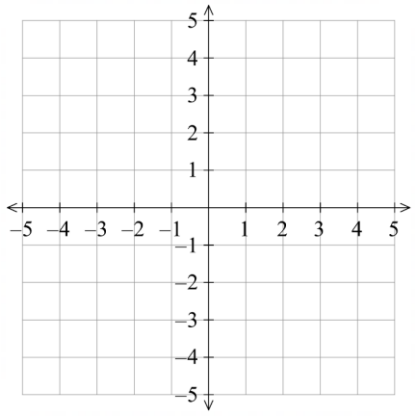
Rise:.....

Run: .....

$m =$ .....

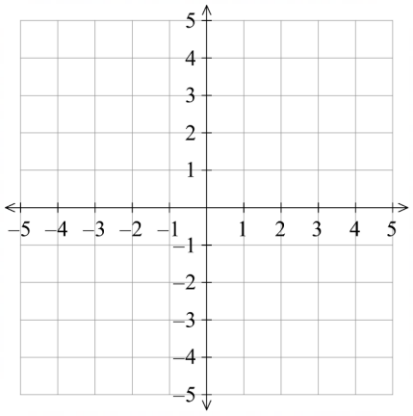
-1

c $A(-1,0)$ and $B(0,-3)$



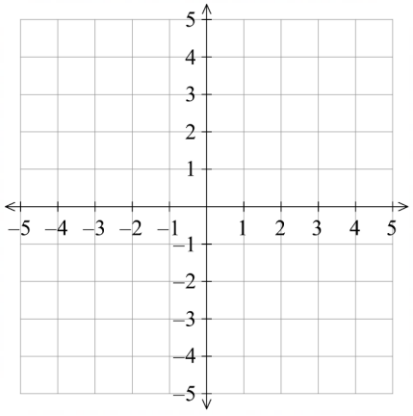
-3

d $A(-2,5)$ and $B(0,2)$



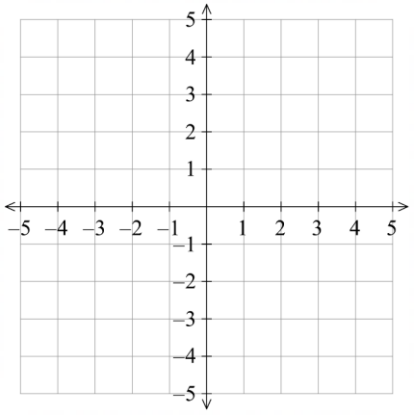
$-\frac{3}{2}$

e $C(-2,4)$ and $D(1,-2)$



-2

f $C(3,-4)$ and $D(1,1)$



$-\frac{5}{2}$

2 Charlotte and Naomi are arguing about a gradient of $\frac{1}{2}$



This means the line rises 1 unit for every 2 units run.

This means the line rises half a unit for every 1 unit run.

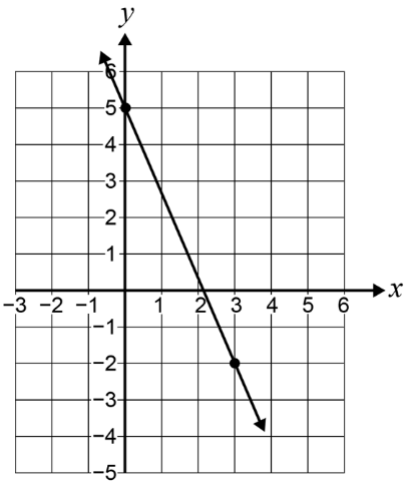


Explain why they are both correct.

Both represent the same ratio of $\frac{1}{2}$

3 **HSC Standard 2 Sample Question Band 4**

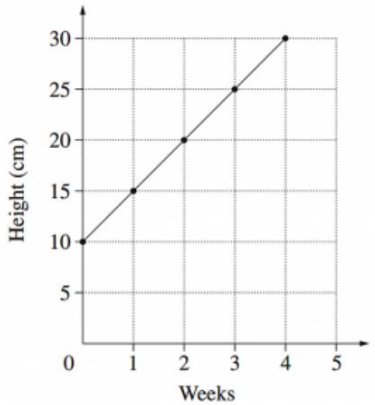
Leo drew a straight line through the points (0, 5) and (3, −2).
What is the gradient of the line?



$m = -\frac{7}{3}$

4 **2004 HSC Standard 2 Band 3**

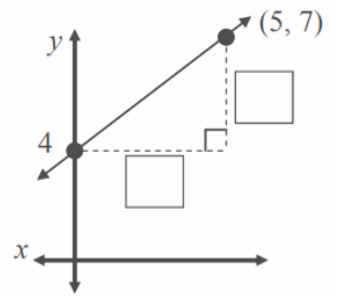
What is the gradient of the line?
The axis scales are different. **Do not count the squares.**



5

5 Calculate the gradient of these lines. There is no grid so you must use the coordinates.

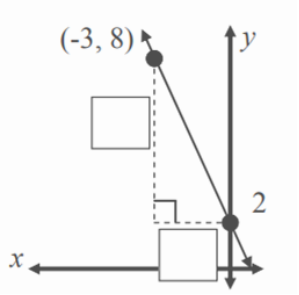
a



$m =$

$\frac{3}{5}$

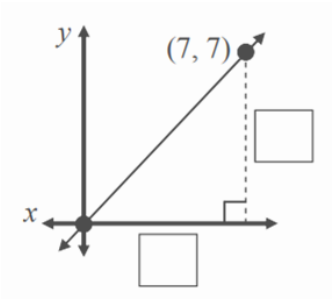
b



$m =$

-2

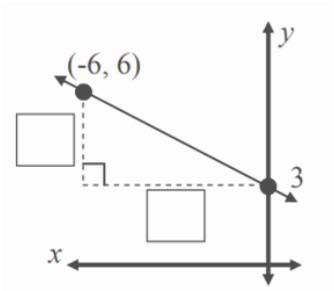
c



$m =$

1

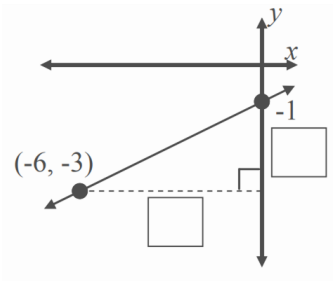
d



$m =$

$-\frac{1}{2}$

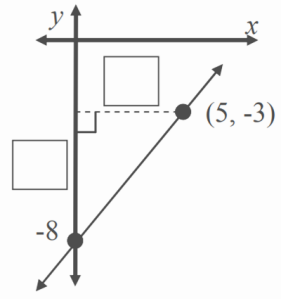
e



$m =$

$\frac{1}{3}$

f

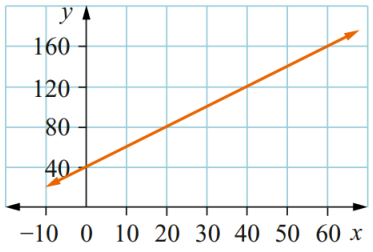


$m =$

1

6 Calculate the gradient of these lines. The axis scales mean **you cannot count the squares**.

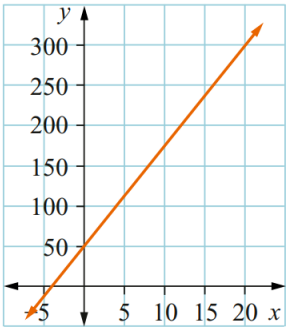
a



$m =$

2

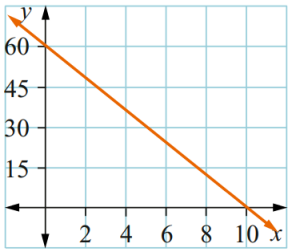
b



$m =$

12.5

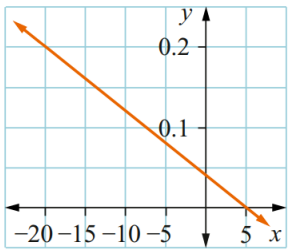
c



$m =$

-6

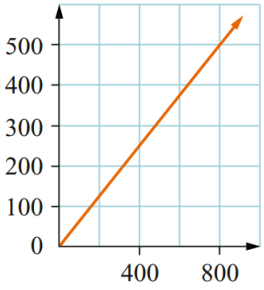
d



$m =$

-0.008

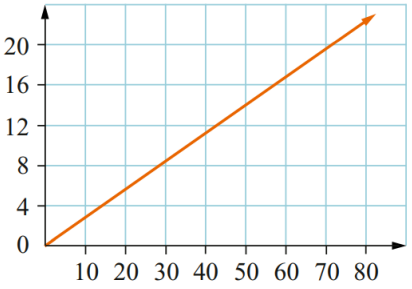
e



$m =$

$\frac{5}{8}$

f

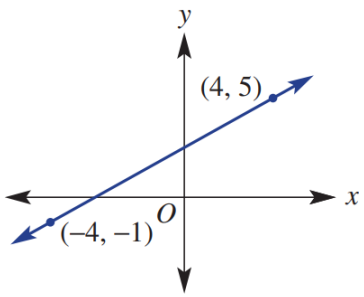


$m =$

$\frac{2}{7}$

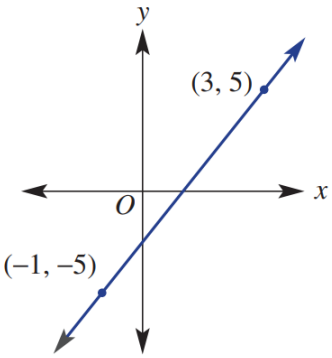
7 Find the gradient of these lines.

a



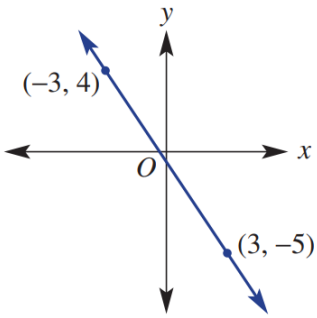
$\frac{3}{4}$

b



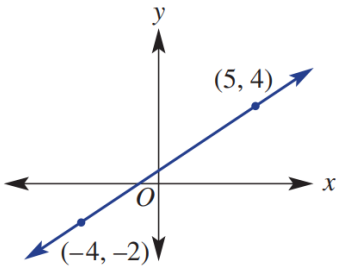
$\frac{5}{2}$

c



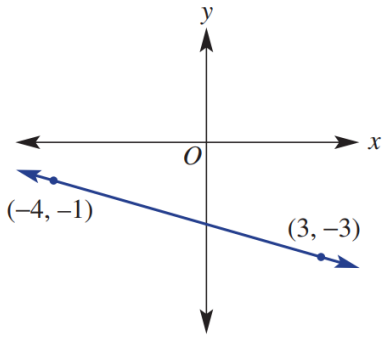
$-\frac{3}{2}$

d



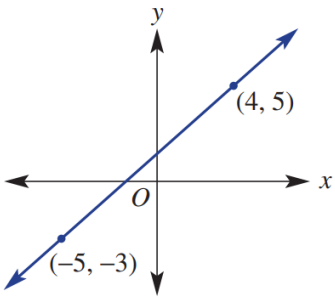
$\frac{2}{3}$

e



$-\frac{2}{7}$

f



$\frac{8}{9}$

g $(-3,1)$ and $(4,-2)$

$-\frac{3}{7}$

h $(-2,-5)$ and $(1,7)$

4

i $(5,-2)$ and $(-4,6)$

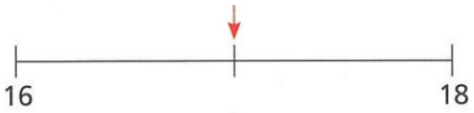
$-\frac{8}{9}$

Midpoint

Review

☐ Identify the midpoint of two numbers

a



$$\frac{\square + \square}{2} = \square$$

b



$$\frac{\square + \square}{2} = \square$$

c



d



e



f



☐ Identify the midpoint of numbers using the mean

a 3 and 5

b 4 and 6

c 4 and 7

d 3 and 7

e 3 and 9

f 3.5 and 9

g 7 and 18

h 7 and 17

i 7 and 16.5

j 7 and 15

k 0.7 and 1.5

l -7 and -15

m -7 and -16

n -6 and -16

o -6 and 16

p 6 and -16

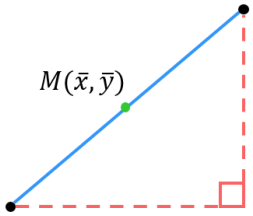
Midpoint Formula

Midpoint is halfway between two endpoints.

Midpoint: https://www.geogebra.org/m/uepqnx7k

To find the midpoint, calculate the average, or mean, of the endpoints: $\text{mean} = \frac{a+b}{2}$

$M = (\text{Mean of } x, \text{Mean of } y)$
$= (\bar{x}, \bar{y})$

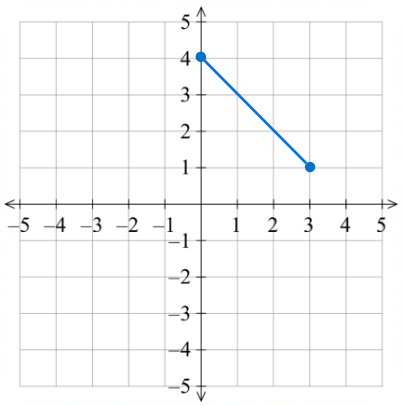


To find the midpoint of an interval:

1. Find x midpoint, $\bar{x}$, the mean of the x-coordinate endpoints.
2. Find y midpoint, $\bar{y}$, the mean of the y-coordinate endpoints.
3. The midpoint is $M = (\bar{x}, \bar{y})$

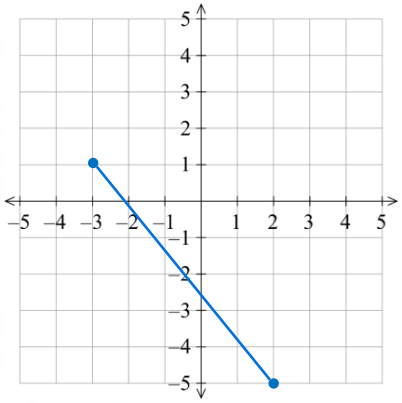
Example Calculate midpoint of an interval

Find the midpoint of $(0,4)$ and $(3,1)$



1. $\bar{x} = \frac{0+3}{2}$
   $= 1.5$
2. $\bar{y} = \frac{4+1}{2}$
   $= 2.5$
3. $M = (1.5, 2.5)$

Find the midpoint of $(-3,1)$ and $(2,-5)$

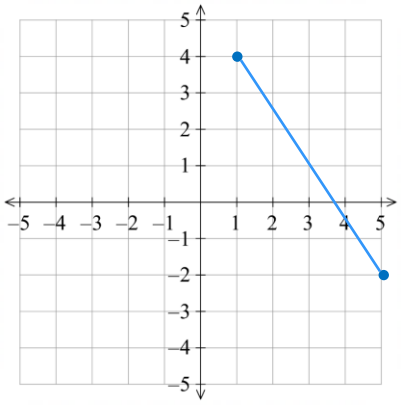


1. $\bar{x} = \frac{-3+2}{2}$
   $= -0.5$
2. $\bar{y} = \frac{1+(-5)}{2}$
   $= -2$
3. $M = (3, 2.5)$

Guided Practice

Find the midpoint of:

a $(5,-2)$ and $(1,4)$



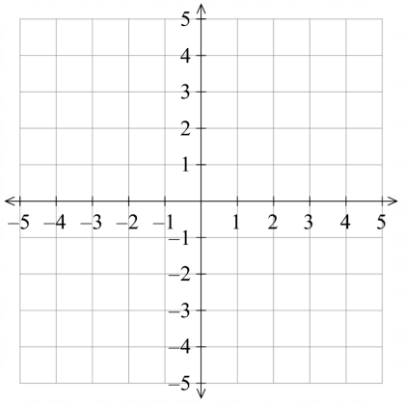
$\bar{x} =$ .....

$\bar{y} =$ .....

$M = (\quad , \quad)$

(3,1)

b $(-2,-5)$ and $(3,-1)$



$\bar{x} =$ .....

$\bar{y} =$ .....

$M = (\quad , \quad)$

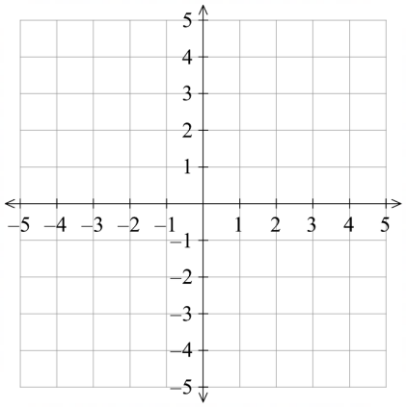
(0.5, −3)

Key Ideas

1. The midpoint is the ..... of the x and y coordinates.

1 Sketch the line interval and calculate the midpoint.

a $(1, 2)$ and $(1, 4)$



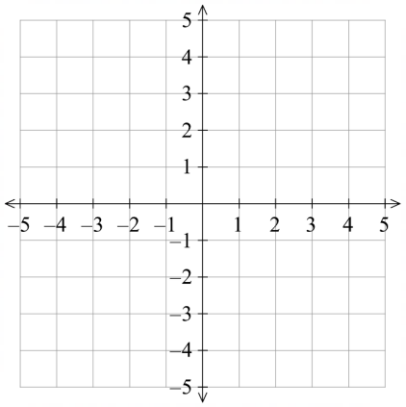
$\bar{x} =$

$\bar{y} =$

$M = (\quad , \quad)$

(1, 3)

b $(-2, 1)$ and $(-2, 5)$



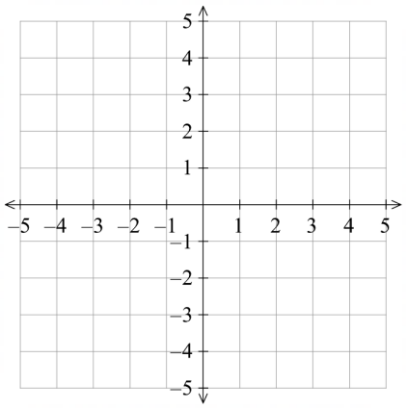
$\bar{x} =$

$\bar{y} =$

$M = (\quad , \quad)$

(−2, 3)

c $(0, 0)$ and $(3, 0)$



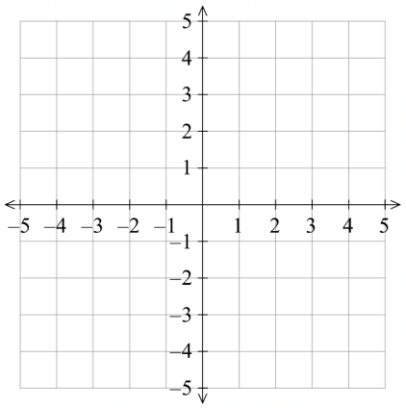
$\bar{x} =$

$\bar{y} =$

$M = (\quad , \quad)$

(1.5, 0)

d $(5, 1)$ and $(3, 1)$



$\bar{x} =$

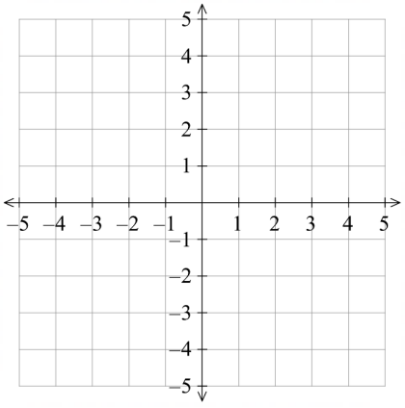
$\bar{y} =$

$M = (\quad , \quad)$

(4, 1)

2

a $(5, 2)$ and $(1, 4)$



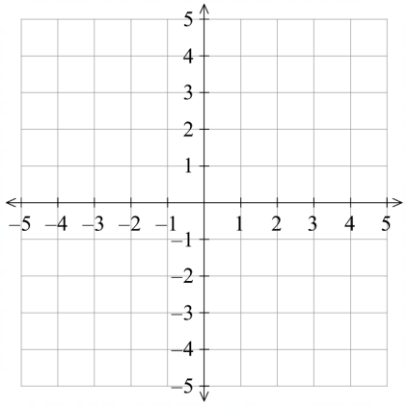
$\bar{x} =$

$\bar{y} =$

$M = (\quad , \quad)$

(3, 3)

b $(2, 0)$ and $(0, 8)$



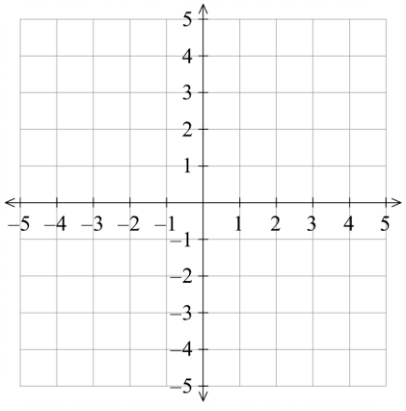
$\bar{x} =$

$\bar{y} =$

$M = (\quad , \quad)$

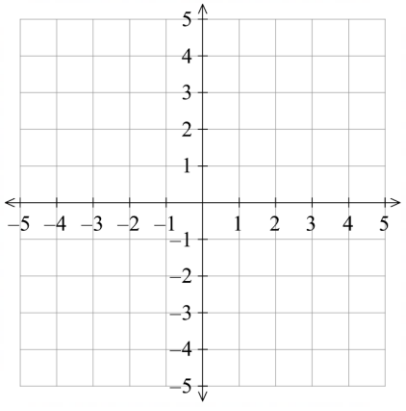
(1, 4)

c $(2, -1)$ and $(-1, 3)$



(0.5, 1)

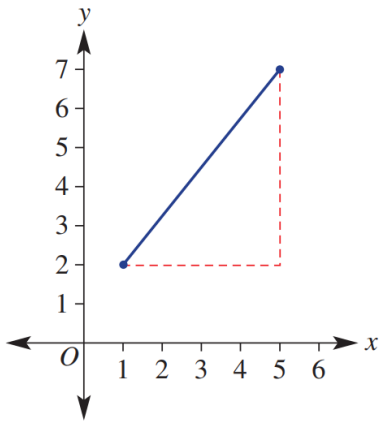
d $(-4, -4)$ and $(-1, 1)$



(−2.5, −1.5)

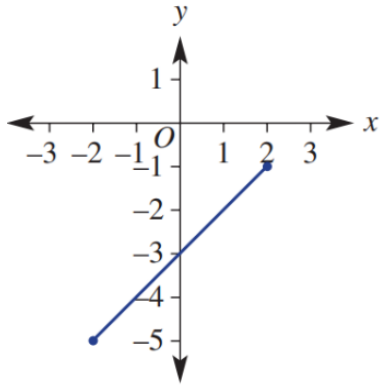
3 Calculate the midpoint of the line segment.

a



$(3, 4.5)$

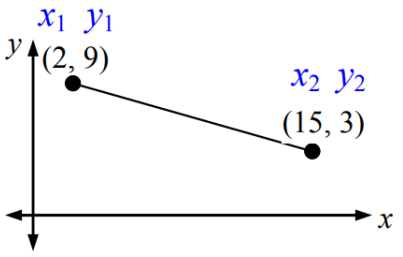
b



$(0, -3)$

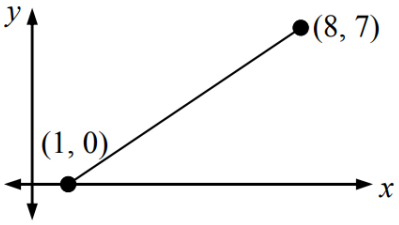
4 Find the midpoint of these line segments.

a



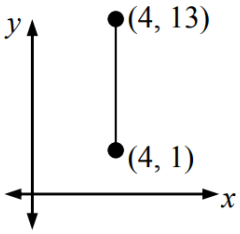
$(8.5, 6)$

b



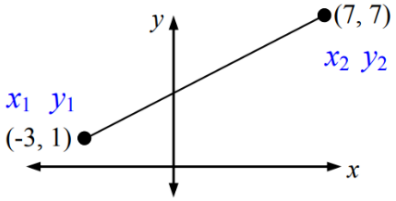
$(4.5, 3.5)$

c



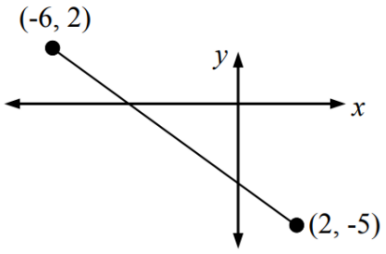
$(4, 7)$

d



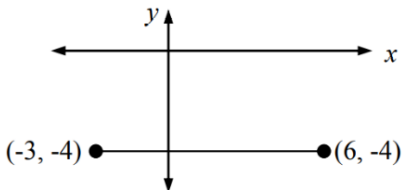
$(2, 4)$

e



$(-2, -1.5)$

f



$(1.5, -4)$

g $(-3,1)$ and $(4, -2)$

$(0.5, -0.5)$

h $(-2, -5)$ and $(1,7)$

$(-0.5, 1)$

i $(5, -2)$ and $(-4,6)$

$(0.5, 2)$

Distance

Review

☐ Calculate square roots and round to 2 d.p.

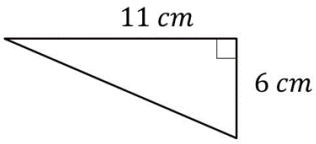
a	$\sqrt{43}$	**b**	$\sqrt{75}$	**c**	$\sqrt{66}$	**d**	$\sqrt{59}$	**e**	$\sqrt{28}$

☐ Calculate squares

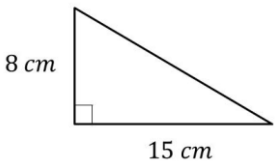
a	$8^2 + 4^2$	**b**	$5^2 + 7^2$	**c**	$6^2 + 3^2$	**d**	$2^2 + 8^2$	**e**	$4^2 + 10^2$

☐ Find the length of the hypotenuse using Pythagoras' theorem, round to 2 d.p.

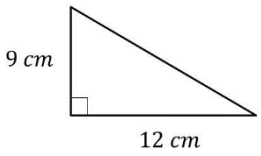
a



b



c



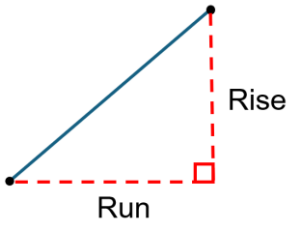
Distance Formula

Distance: https://www.geogebra.org/m/t94gke48

Pythagoras' theorem calculates the distance, or length, between two points.

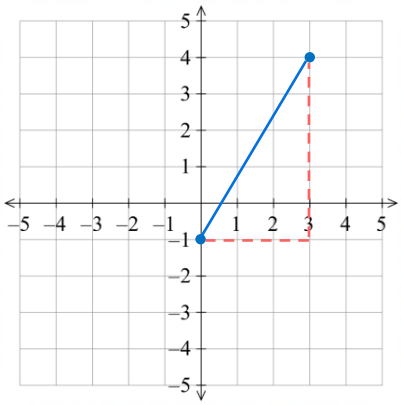
$$d^2 = \text{rise}^2 + \text{run}^2$$

$$d = \sqrt{\text{rise}^2 + \text{run}^2}$$



Example Calculate length of an interval

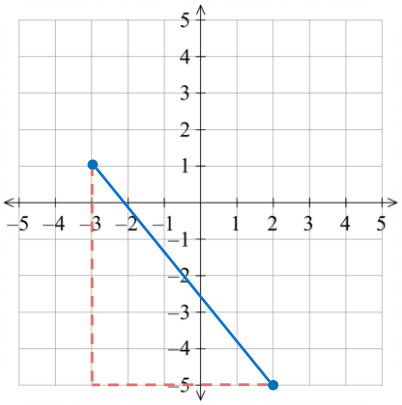
Find the length between $(0,-1)$ and $(3,4)$



$d = \sqrt{5^2 + 3^2} = \sqrt{34}$

≈ 5.83 units

Find the length between $(-3,1)$ and $(2,-5)$



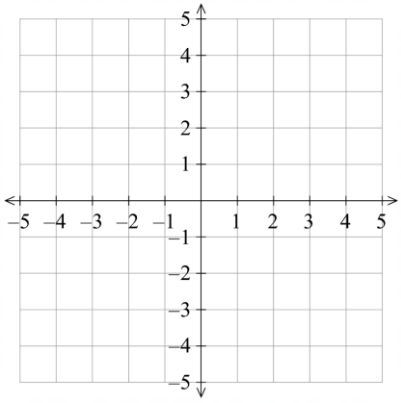
$d = \sqrt{(-6)^2 + 5^2} = \sqrt{61}$

≈ 7.81 units

Guided Practice

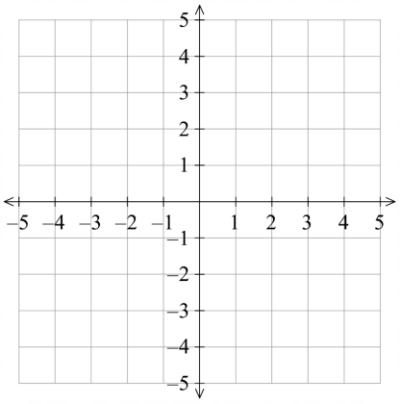
Find the length, to 2 d.p. where necessary, between:

a $(1,1)$ and $(5,4)$



5

b $(-4,-3)$ and $(2,1)$



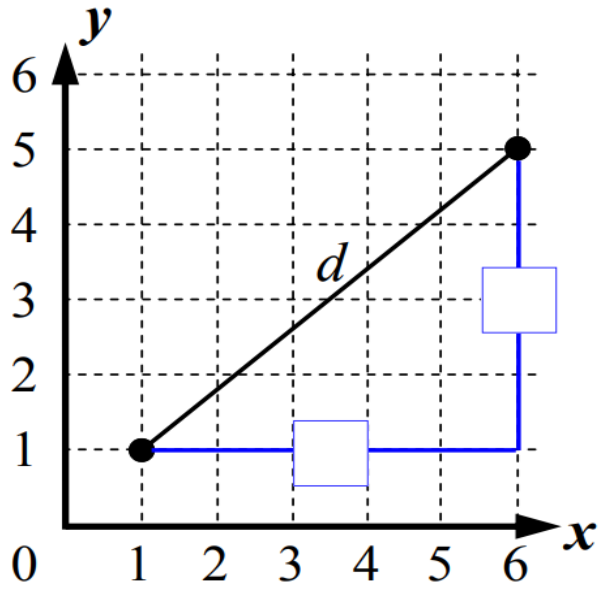
7.21

Key Ideas

1. The length between two coordinates is calculated using …… theorem.

1 Calculate the distance of these intervals. Check your answer by measuring with a ruler.

a



Rise ……

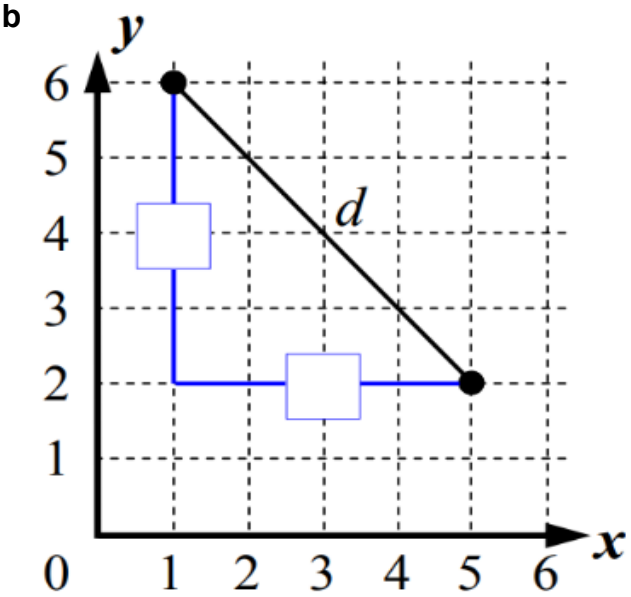
Run ……

$d = \sqrt{(\quad)^2 + (\quad)^2}$

$=$

6.40

b



Rise = ……

Run = ……

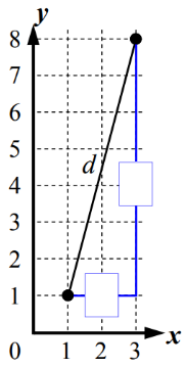
$d = \sqrt{(\quad)^2 + (\quad)^2}$

$=$

5.66

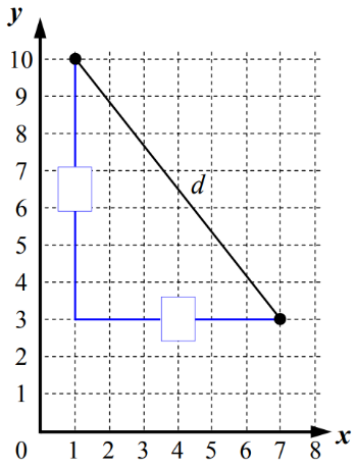
2 Calculate the distance of these intervals to 2 d.p. You won't be able to use a ruler.

a



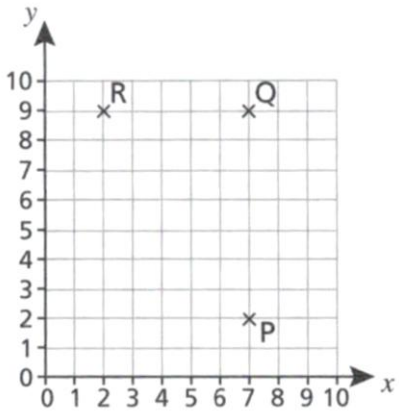
7.28

b



9.22

3



a Write the length of PQ

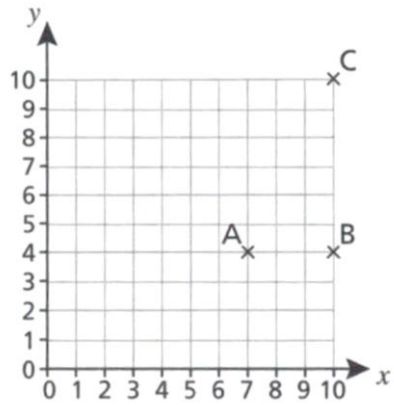
7 units

b Write the length of QR

5 units

c Use Pythagoras' theorem to show that PR = 8.6 units to 1 decimal place.

4



a Write the length of AB

3 units

b Write the length of BC

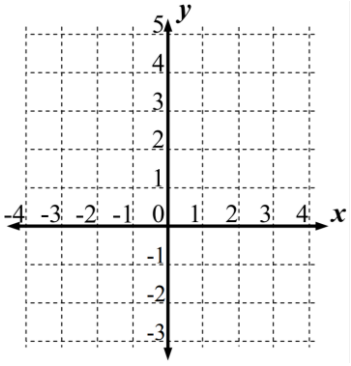
6 units

c Use Pythagoras' theorem to work out the length of AC. Give your answer to 1 decimal place.

6.7 units

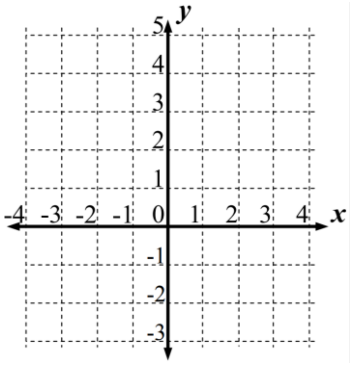
5 Calculate the distance of these intervals. Answer to 1 d.p.

a $(4,5)$ and $(1,-3)$



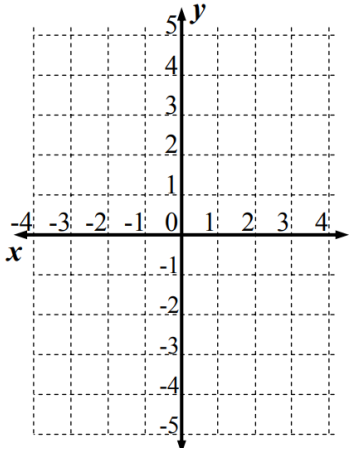
8.5

b $(-3,4)$ and $(-1,-2)$



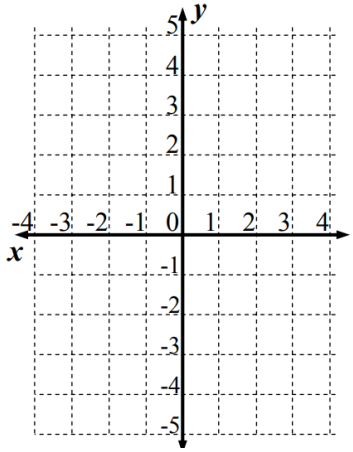
6.3

c $(-4,4)$ and $(2,5)$



6.1

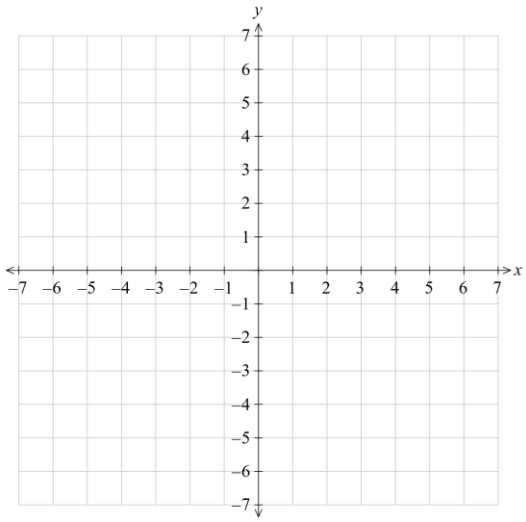
d $(-4,2)$ and $(1,-5)$



8.6

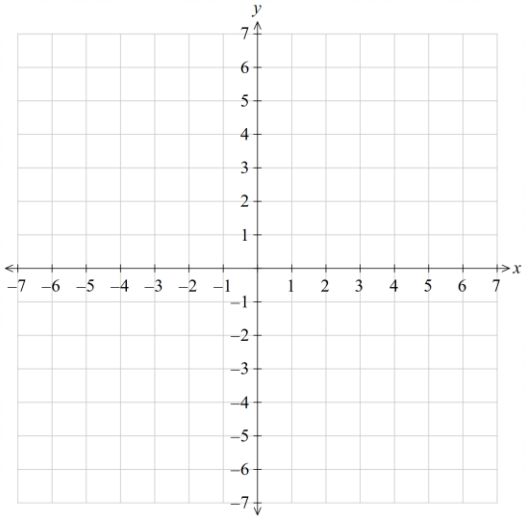
6 Plot these intervals on the Cartesian plane and calculate the distance to 1 d.p.

a $(0, 1)$ and $(4, 7)$



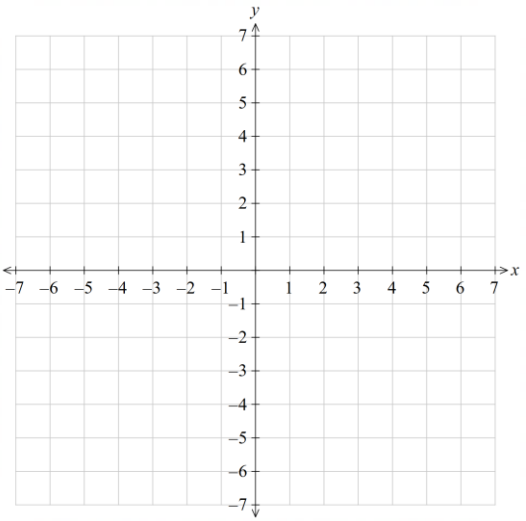
7.2

b $(-2, -1)$ and $(4, -2)$



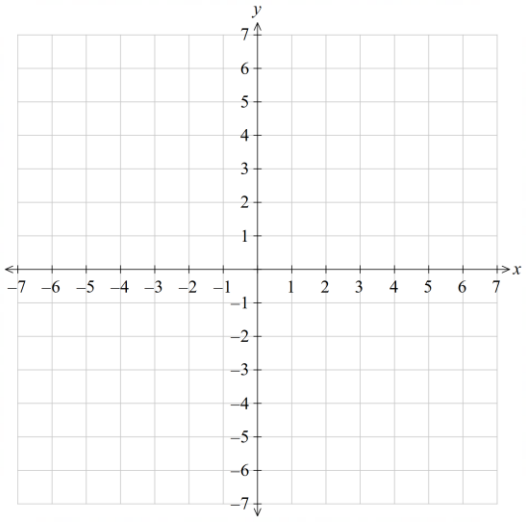
6.1

c $(-5, 3)$ and $(6, 2)$



11.0

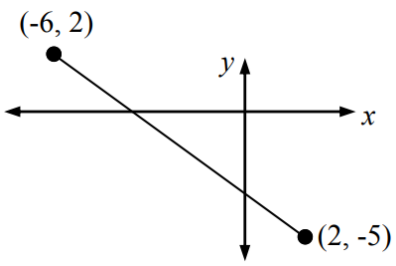
d $(-7, 0)$ and $(5, -4)$



12.6

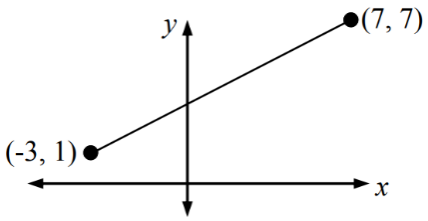
7 Calculate the distance between these two points to 1 d.p.

a



10.6

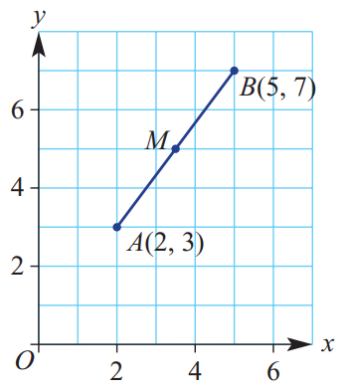
b



11.7

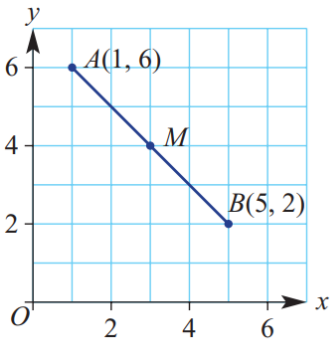
8 Find the midpoint, gradient, and length of these line segments. Give lengths to 1 d.p.

a



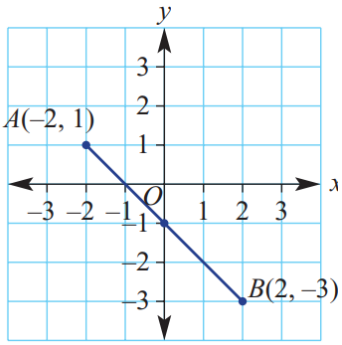
Midpoint: (3.5,5), Gradient: $\frac{4}{3}$, Length: 5.0

b



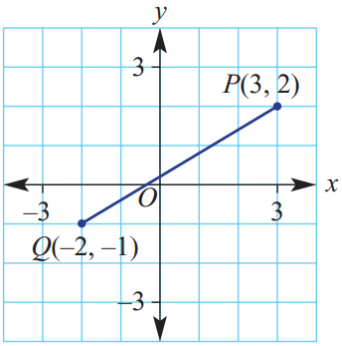
Midpoint: (3,4), Gradient: -1, Length: 5.7

c



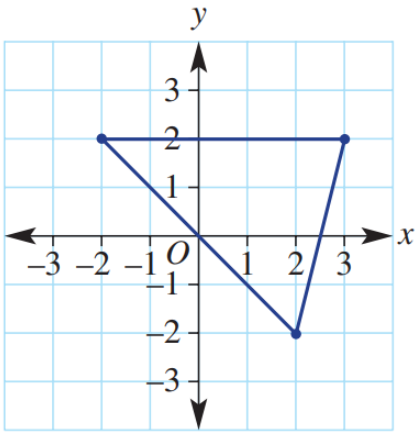
Midpoint: $(0,-1)$, Gradient: -1, Length: 5.7

d



Midpoint: $\left(\frac{1}{2},\frac{1}{2}\right)$, Gradient: $\frac{3}{5}$, Length: 5.8

9 A block of land is shown on this map. Each unit distance represents 100 metres in real life. Calculate the perimeter of the block.



1478 m